

AI-POWERED INTELLIGENT SALES FORECASTING PLATFORM USING PREDICTIVE ANALYTICS GROUP-1

Team Members & Roles:

No.	Team Member	Role
1	Ritika Kushwaha	Project Lead / Scrum Master, Backend Developer
2	Aditi Bhosure	Database / DevOps Engineer
3	Samraiyam Nallagorla	AI/ML Engineer
4	Tejashwini M	Frontend Developer
5	Rushikesh Arepalli	QA / Test Engineer

Project Overview:

SalesGenie AI is an AI-based sales assistant designed to help sales teams manage leads and better understand potential customers. It brings lead management, lead analysis, lead scoring, AI-powered outreach, recommendations, CRM integration, and sales analytics together in one platform.

The project reduces manual work by using available lead and conversation information to provide useful insights and suggest appropriate next actions. Each module contributes to the overall sales workflow.

Project Objective:

The main objective of SalesGenie AI is to create an intelligent sales assistant that helps sales teams manage leads in a simpler, faster, and more organized way. The platform is designed to help teams:

- Manage and organize leads in one place.
- Analyze lead and company details using AI.
- Create personalized outreach messages.
- Generate lead qualification scores and recommendations.
- Connect CRM data and activity information.
- Use conversation summaries and sales insights to support follow-up actions.

Project Modules:

The SalesGenie AI project is divided into six main modules. Each module focuses on a different part of the sales process, and together they form the complete platform.

Module 1 - Lead Management & Prospect Database

Manages lead and prospect information in one place.

Module 2 - Lead Intelligence & Company Analysis

Uses AI to provide insights from lead and company information.

Module 3 - AI Outreach Generation

Creates AI-generated emails and outreach content tailored to lead information and industry.

Module 4 - Lead Scoring & Recommendation Engine

Provides lead scores and recommends follow-up timing, channel, and content strategy.

Module 5 - Conversation Intelligence & CRM Integration

Connects the platform with CRM workflows and provides conversation summaries, action items, sync status, and activity information.

Module 6 - Dashboard & Sales Analytics

Provides a central view of sales information, insights, and follow-up activity.

Milestone 3 Delivery Summary:

Milestone	Duration	Development Period	QA / Review	Focus Module	Acceptance Focus	Status
Milestone 3	Day 21-25	12-Aug-2026 to 17-Aug-2026	18-Aug-2026	Module 5: CRM Integration & Conversation Intelligence	CRM workflows operate correctly; conversation summaries are generated successfully	Signed Off

Milestone - 3: QA Report

Testing Overview:

Milestone 3 was the third major testing checkpoint of the SalesGenie AI project. The focus was to verify the Conversation Intelligence and CRM Integration features before moving to the next phase of development.

The milestone covered CRM integration APIs with Salesforce/HubSpot schema mapping, CRM sync logic for contacts, activities, and deal stages, sync logs, meeting and call summarization using an LLM, and the Streamlit Conversations tab with CRM sync status, meeting summaries, and an activity feed.

QA activities included functional testing, CRM sync workflow testing, mapping validation, summary-output review, integration testing, and end-to-end testing. The completed workflow was reviewed on Day 25 and Milestone 3 was signed off.

Testing Objectives:

The main goal of Milestone 3 testing was to make sure that CRM data could be integrated and synchronized correctly and that conversation summaries could support the sales workflow. Testing focused on the following areas:

1. To verify the CRM integration API endpoints and Salesforce/HubSpot schema mapping.
2. To check CRM synchronization for contacts, activities, deal stages, and sync logs.
3. To verify that the Conversations tab displayed CRM sync status and activity information correctly.
4. To validate meeting/call summaries for key points and extracted action items.
5. To review whether the generated summary was relevant to the supplied conversation content.
6. To make sure that CRM, backend, AI summarization, and the Streamlit interface communicated correctly.
7. To identify, resolve, and retest functional or integration issues before milestone sign-off.

Testing Scope:

The testing scope for Milestone 3 was mainly divided into the following areas.

CRM Integration APIs

- Testing CRM endpoints and supported Salesforce/HubSpot schema mapping.
- Checking request handling, response data, and mapped field availability.

CRM Sync Service

- Synchronizing contact information, activities, and deal stages.
- Checking sync-log creation and status visibility.
- Reviewing retry behavior for a failed or delayed sync attempt.

Conversation Intelligence

- Generating meeting/call summaries.
- Reviewing extraction of key points and action items.
- Checking that the generated summary was displayed in the Conversations tab.

Integration

- Checking the Conversations tab for sync status, summary content, and activity feed.
- Validating the complete CRM sync and conversation-summary workflow end to end.

Testing Environment & Tools:

Area	Tool / Technology
API Testing	Postman
Backend	FastAPI
Database	PostgreSQL
ORM	SQLAlchemy
Frontend	Streamlit
AI Integration	Gemini / AI API
Automated Testing	PyTest
Version Control	Git
Manual Testing	Application UI / Browser

Testing Approach:

Milestone 3 used manual, API, functional, integration, and end-to-end testing. CRM workflows were checked through supported mapping and sync scenarios, while meeting/call summaries were reviewed for clear key points and action items. The complete user flow was validated through the Conversations tab.

Test Cases:

Test ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-20	CRM Integration	Verify CRM API endpoint and schema mapping	Mapped CRM fields should be accepted and returned correctly	CRM mapping endpoint responded with expected data	Pass

Test ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-21	CRM Sync	Sync contact information	Contact data should be synchronized and logged	Contact sync was completed and logged	Pass
TC-22	CRM Sync	Sync activities and deal stages	Activities and deal stages should be synchronized	CRM workflow sync was completed	Pass
TC-23	CRM Sync	Review sync status after retry	Latest sync status should be displayed correctly	Status refresh was verified after correction	Pass
TC-24	Conversation Intelligence	Generate meeting/call summary	Key points should be generated from conversation content	Summary was generated successfully	Pass
TC-25	Conversation Intelligence	Extract action items	Clear action items should be extracted and displayed	Action items were generated and reviewed	Pass
TC-26	Conversations UI	Display CRM sync status, summary, and activity feed	Required information should be visible in Conversations tab	UI display was validated	Pass
TC-27	Integration	Run CRM sync through Conversations tab	CRM and UI components should communicate correctly	Integration workflow was validated	Pass
TC-28	End-to-End	Run CRM sync and summary workflow	Sync results and conversation summary should be available end to end	End-to-end workflow was validated	Pass

Test Execution Metrics:

Milestone 3 testing covered the documented CRM integration, sync, conversation-summary, interface, integration, and end-to-end scenarios. Two issues found during the initial execution were resolved and successfully retested before sign-off.

Metric	Result
Total Test Cases	9
Test Cases Executed	9
Passed - Initial Execution	7
Failed - Initial Execution	2
Blocked	0
Initial Pass Percentage	77.78%
Defects Identified	2
Defects Resolved	2
Defects Retested	2
Final Passed After Retesting	9
Final Pass Percentage	100%

Defect Summary:

During the initial Milestone 3 QA execution, two medium-severity issues were identified. Both affected the visibility or readability of CRM and conversation information but did not block the overall application. The issues were shared with the respective team members, corrected, and verified through retesting.

Defect ID	Module	Issue	Severity	Status
BUG-03	CRM Sync	Sync status did not refresh immediately after a retry.	Medium	Resolved
BUG-04	Conversation Intelligence	Action items in a multi-line summary were not displayed clearly in the Conversations tab.	Medium	Resolved

Severity

- Critical: Major functionality is completely blocked.
- High: Important functionality is seriously affected.
- Medium: A specific feature is affected but the application can still be used.
- Low: Minor UI or usability issue.

Both defects were classified as Medium because they affected the CRM and summary experience but did not stop the complete workflow from functioning.

Defect Resolution & Retesting:

The identified issues were discussed with the relevant team members and corrected. After the fixes were completed, the same scenarios were executed again to confirm that the sync status and action-item display were working as expected.

Defect ID	Issue	Retest Result	Final Status
BUG-03	CRM sync status refresh after retry	Passed	Closed
BUG-04	Action-item display in conversation summary	Passed	Closed

Summary

Both identified defects were fixed and successfully verified through retesting. After the fixes, all documented Milestone 3 test scenarios passed successfully, allowing the milestone to proceed to completion.

Integration Testing:

Integration testing was performed to verify that CRM integration APIs, sync logic, the AI summarization service, and the Conversations UI worked together properly. The complete flow checked was:

Conversations UI -> FastAPI Backend -> CRM Sync / AI Summary Service -> Backend -> Conversations UI

Areas Checked

- Whether CRM requests were received and mapped correctly by the backend.
- Whether contacts, activities, and deal stages could be synchronized and logged.
- Whether sync status was returned correctly after synchronization or retry.
- Whether the AI summarization service received the conversation content and returned key points and action items.

- Whether the Conversations tab displayed sync status, summary, and activity feed information correctly.

CRM Integration / Conversation Intelligence Validation:

Milestone 3 validation focused on whether CRM information could move through the supported workflow and whether conversation summaries were useful and available in the application.

CRM Integration Validation

- Salesforce/HubSpot schema mapping was included in API validation.
- CRM sync logic for contacts, activities, deal stages, and sync logs was reviewed.
- The retry-related status refresh issue was corrected and retested successfully.

Conversation Intelligence Validation

- Meeting and call content was used to generate summaries with key points.
- Action items were extracted and reviewed in the Conversations tab.
- The multi-line action-item display issue was corrected and retested successfully.

QA Observation

The QA review focused on both feature availability and workflow usefulness: whether CRM data could be synchronized and tracked, and whether the summary output clearly captured key points and action items for follow-up. Functional, integration, and end-to-end validation were included in the Day 25 review.

Overall QA Results:

Testing Area	Result
CRM API and Schema Mapping	Passed
CRM Contact / Activity / Deal-stage Sync	Passed
CRM Sync Logs and Status	Passed
Conversation Summary Generation	Passed
Key-point and Action-item Extraction	Passed
Conversations Tab Validation	Passed
Integration Testing	Passed
Defect Retesting	Passed
End-to-End Testing	Passed
Milestone 3 QA Review / Sign-off	Completed and Signed Off

Overall QA Summary

Milestone 3 testing covered CRM Integration and Conversation Intelligence features, including CRM API mapping, sync logic, sync logs, meeting/call summarization, action-item extraction, and the Conversations tab. The QA review included functional testing, CRM workflow testing, summary-output review, integration testing, and end-to-end testing.

Two medium-severity issues were identified during initial execution. Both were resolved and successfully retested, and all documented test scenarios passed in the final execution.

Based on the completed validation and Day 25 review, Milestone 3 was approved and signed off.

Conclusion:

Milestone 3 QA was completed to verify that the CRM Integration and Conversation Intelligence features were working properly and communicating correctly with the Conversations interface. The QA activities covered CRM mapping, data synchronization, sync status, conversation summaries, action-item extraction, integration testing, and end-to-end testing.

The completed review confirmed that CRM workflows and sales conversation summaries had been validated for the milestone. The issues identified during testing were resolved and verified through retesting. Overall, Milestone 3 was completed and signed off, providing a validated foundation for the remaining SalesGenie AI features.

Milestone Sign-off:

Milestone	QA / Review Date	Validation Coverage	Final Status
Milestone 3 - CRM Integration & Conversation Intelligence	18-Aug-2026	CRM workflow, summary-output, integration, defect retesting, and end-to-end testing	Completed and Signed Off